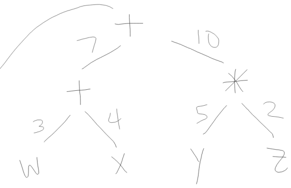
Use int w=3, x=4, y=5, z=2; for #1 – 5.

1. (w + x) + (y * z)

1 3 2 7 4 6 5

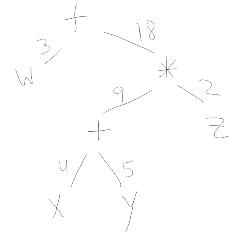


17

Use int w=3, x=4, y=5, z=2; for #1 – 5.

2. w + ((x + y) * z)

1 7 2 4 3 6 5



21

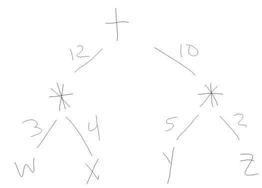
Use int w=3, x=4, y=5, z=2; for #1 – 5.

4. w + z – Math.pow(z++, y – z);

Use int w=3, x=4, y=5, z=2; for #1 – 5.

3. (w * x) + (y * z)

1 3 2 7 4 6 5

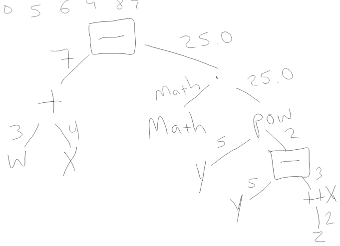


22

Use int w=3, x=4, y=5, z=2; for #1 – 5.

4. (w + x) – Math.pow(y,y – ++z);

1 3 2 12 4 11 10 5 6 9 8 7

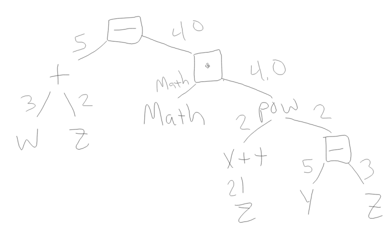


-18.0

Use int w=3, x=4, y=5, z=2; for #1 – 5.

4. w + z – Math.pow(z++, y – z);

1 3 2 12 4 11 10 5 6 7 9 8

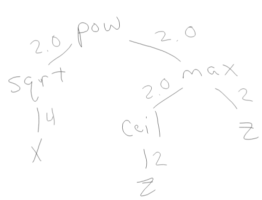


1.0

Use int w=3, x=4, y=5, z=2; for #1 – 5.

5. pow(sqrt(x) , max(ceil(z),z));

7 2 1 6 4 3 5

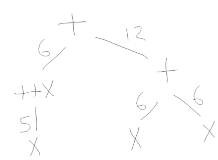


4.0

Use int x=5; for #6 – 16.

6. ++x + (x + x)

2 1 6 3 5 4

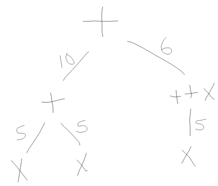


18

Use int x=5; for #6 – 16.

7. (x + x) + ++x

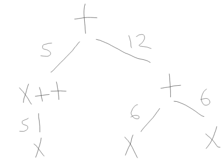
1 3 2 6 5 4



16

8. x++ + (x + x)

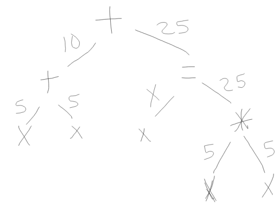
1 2 6 3 5 4



17

9. (x + x) + (x = x * x)

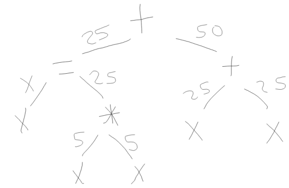
1 3 2 9 4 8 5 7 6



35

10. (x = x * x) + (x + x)   X = ~~5~~ 25

1 5 2 4 3 9 6 8 7



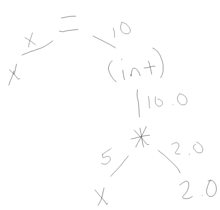
75

Use int x=5; for #6 – 16.

11. x *= 2.0;

X = (int)(X * 2.0)

1 6 5 2 4 3



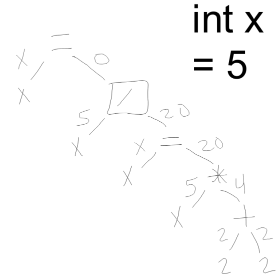
10

12. x/=x*=2+2;

int x = 5

x = x / (x*=2+2)

x = x / (x = x * (2+2))

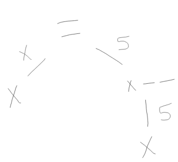


X=0

13. x = x--;

1 4 2 3

int x = 5

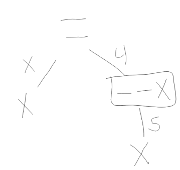


X=5

14. x = --x;

1 4 3 2

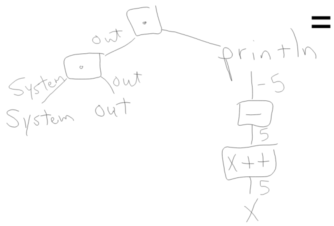
int x = 5



X=4

```
15. System.out.println(-x++);
```

int x = 5

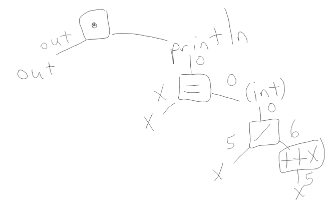


-5

```
16. out.println(x/=++x);
```

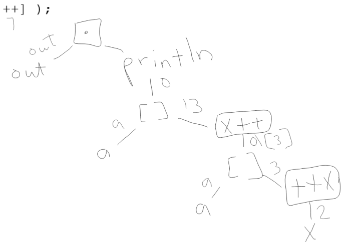
int x = 5

out.println(x = (int) (x / ++x))



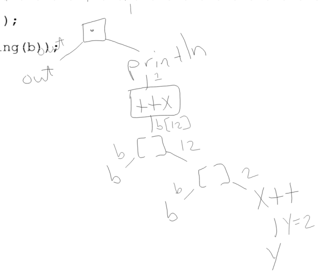
X=0

```
17. int x = 2;
    int[] a = {10,11,12,13,14,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0};

    out.println( a[a[++x]++] );
```



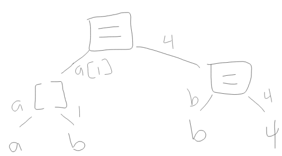
0

```
18. int y=2;
    int[] b = {10,11,12,13,14,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0};

    out.println( ++b[b[y++]] );

    out.println(Arrays.toString(b));
```



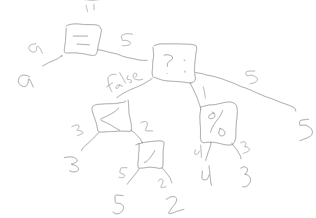
1

```
19. int[] a = new int[5];
    int b = 1;
    a[b] = b = 4;

    out.println(Arrays.toString(a));
```



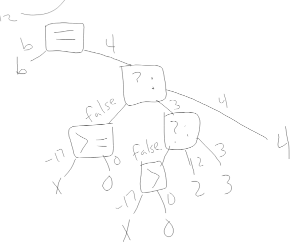
[0,4,0,0,0]

```
20. int a = (3 < (5 / 2) ? (4 % 3) : 5;
```



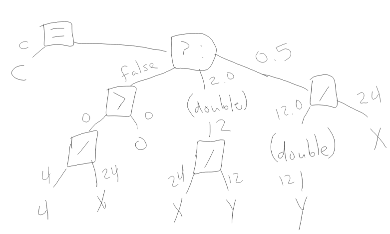
a=5

```
21. int x = -17;
    int b = (x>=0) ? (x>0) ? 2 : 3) : 4;
```



b=4

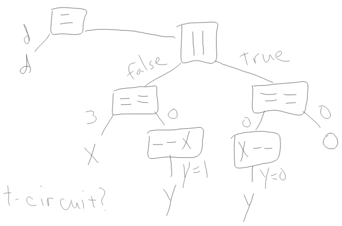
```
22. int x = 24, y = 12; double c;
    c = (4/x > 0) ? (double)(x/y) : (double)y/x;
```



0.5

```
23. int x = 3, y = 1; boolean d;
    d = (x == --y) || (y-- == 0);
```

1 11 2 5 4 3 10 6 7 9 8



d=true

Did the OR short-circuit?

Note that you must give the instanceof an object reference!

- `String s;`
- `s instanceof Object`
  `java.lang.RuntimeException: variable s has not been assigned a value`
- `s="Fine"`
- `s instanceof Object`
  `true`
- `"Fine" instanceof Object`
  `true`
- `"Fine".substring(0,1) instanceof String`
  `true`

```
public class S {

  static int    x  = 2;
  static int[] ray = new int[8];

  public static int[] a() { Arrays.fill(ray,1);  return ray;
  public static int[] b() { Arrays.fill(ray,2);  return ray;
  public static int[] c() { Arrays.fill(ray,3);  return ray;
  public static int[] d() { Arrays.fill(ray,4);  return ray;

  public static int p() { return x=2; }
  public static int q() { return x=3; }
  public static int r() { return x=5; }
  public static int s() { return x=7; }
```

```
24)  public static void main(String[] args) {
       a()[p()] = q();
```

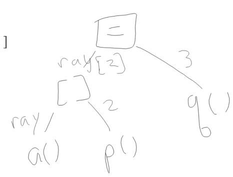
```
(new int[10])[0]   -->   0
 new int[10][0]   -->   [[I@53e25b76
(new int[] {3,1,4})[2]   -->   4
```

```
24)  public static void main(String[] args) {
       a()[p()] = q();
       out.println(Arrays.toString(a));
       out.println(x);
     }
```

1 3 2 5 4

a = [1, 1, 3, 1, 1, 1, 1, 1]

x = ~~2~~ 3



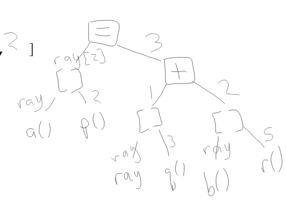

```
public class S {

  static int    x  = 2;
  static int[] ray = new int[8];

  public static int[] a() { Arrays.fill(ray,1);  return ray;
  public static int[] b() { Arrays.fill(ray,2);  return ray;
  public static int[] c() { Arrays.fill(ray,3);  return ray;
  public static int[] d() { Arrays.fill(ray,4);  return ray;

  public static int p() { return x=2; }
  public static int q() { return x=3; }
  public static int r() { return x=5; }
  public static int s() { return x=7; }
```


```
)  public static void main(String[] args) {
     a()[p()] = ray[q()] + b()[r()];
     out.println(Arrays.toString(a));
     out.println(x);
   }
```

1 3 2 11 4 6 5 10 7 9 8

a = [2, 2, 3, 2, 2, 2, 2, 2]

x = ~~2~~ ~~3~~ 5

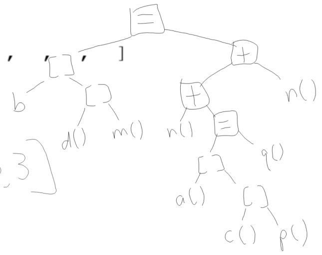



```
public class S {

  static int    x  = 2;
  static int[] ray = new int[8];

  public static int[] a() { Arrays.fill(ray,1);  return ray;
  public static int[] b() { Arrays.fill(ray,2);  return ray;
  public static int[] c() { Arrays.fill(ray,3);  return ray;
  public static int[] d() { Arrays.fill(ray,4);  return ray;

  public static int p() { return x=2; }
  public static int q() { return x=3; }
  public static int r() { return x=5; }
  public static int s() { return x=7; }
```

25)
```
public static void main(String[] args) {
  b()[d()[m()]] = n() + (a()[c()[p()]] = q()) + n();
   1  5 2 4 3  17 6  14  7 11 8 10 9   13 12  16 15
  out.println(Arrays.toString(a));
  out.println(x);
}
```

a = [, , , , , , ,]

x = ___



[3, 3, 3, 7, 13, 3, 3, 3]

x=3

a = [, , , , , , ,]

x = ___